

Finance Metrics Validator – Technical Deep Dive

Architecture Components

- SQL parsing via sqlglot for structural analysis
- Static finance-risk rules engine
- In-memory DuckDB execution for reconciliation
- Pydantic schemas for structured reporting
- CLI entrypoint with exit codes for CI integration

Testing Strategy

- Pytest suite validating static checks
- Reconciliation failure detection
- Alias handling and tolerance testing

Risk Controls Implemented

- Missing status filter detection
- Refund omission detection
- Aggregation integrity checks
- Numeric reconciliation threshold enforcement

Production Readiness

- Editable install via pyproject.toml
- Dependency isolation

- Deterministic outputs
- CI-compatible exit codes

Architecture Diagram Included Separately